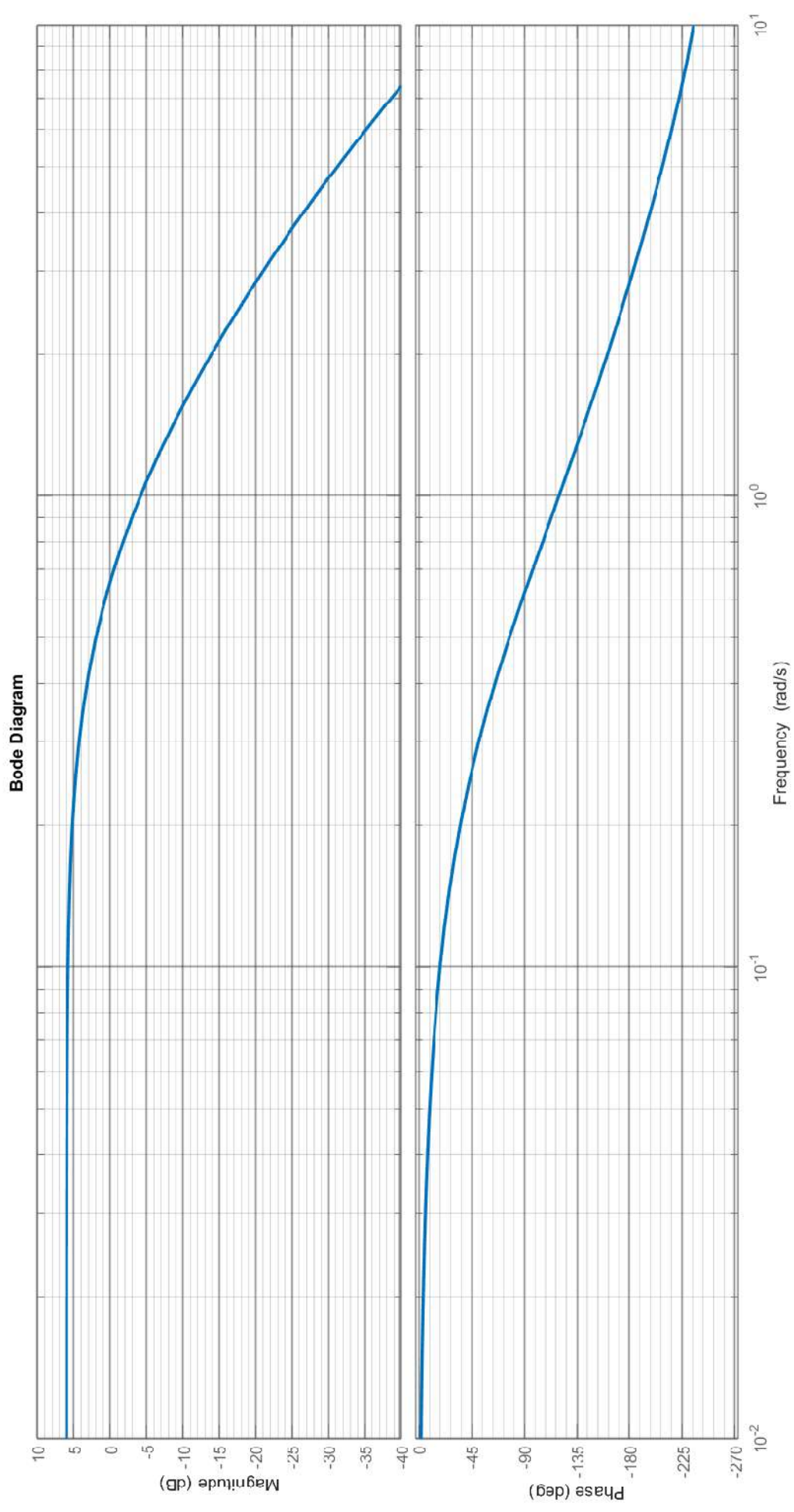
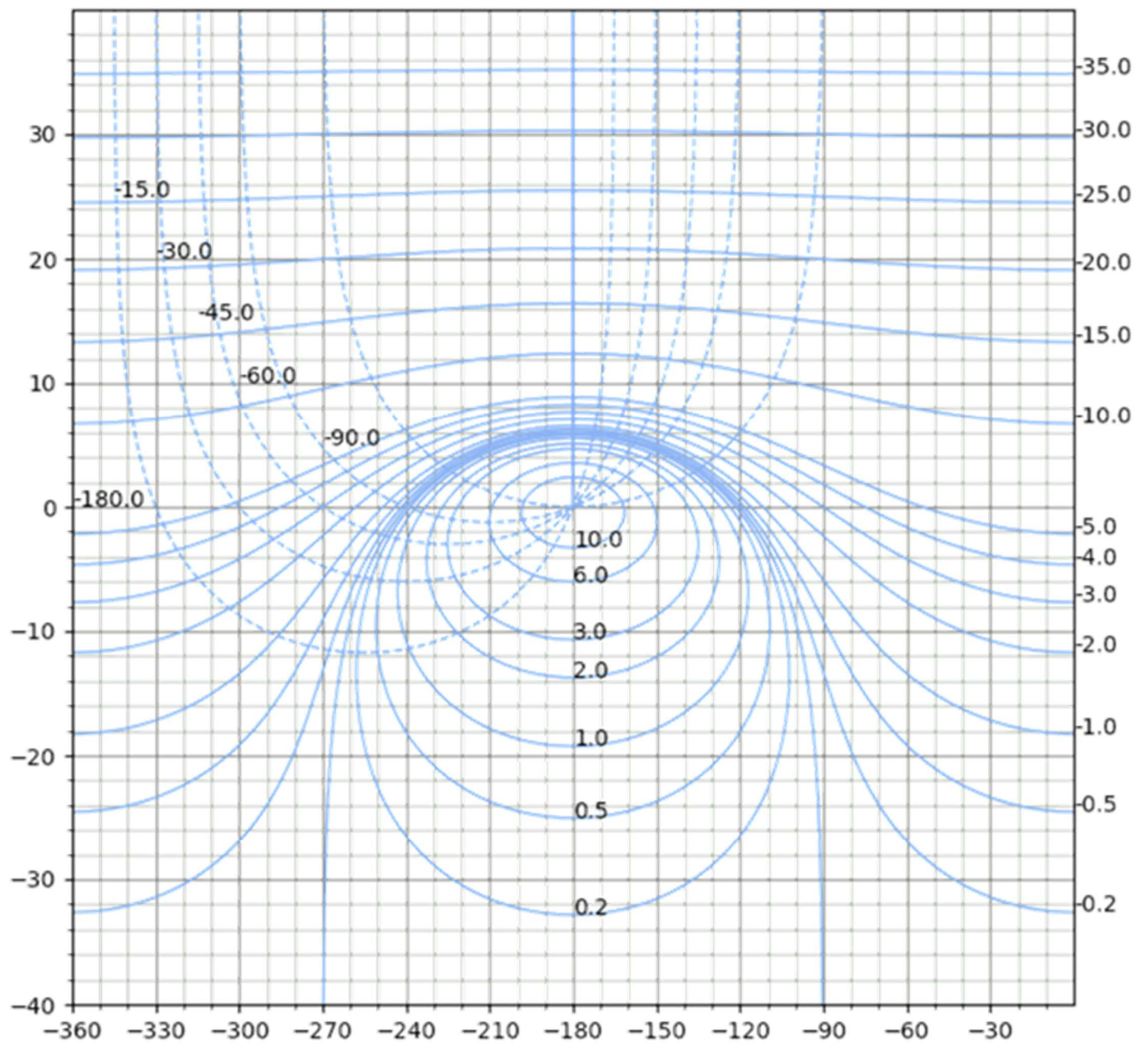


Tut Test 5 [15 marks]

The Bode plot of a system is shown on the next page. The system is stable in open loop, and will be placed in negative feedback with a proportional controller with the aim of improving its performance to meet the following specifications:

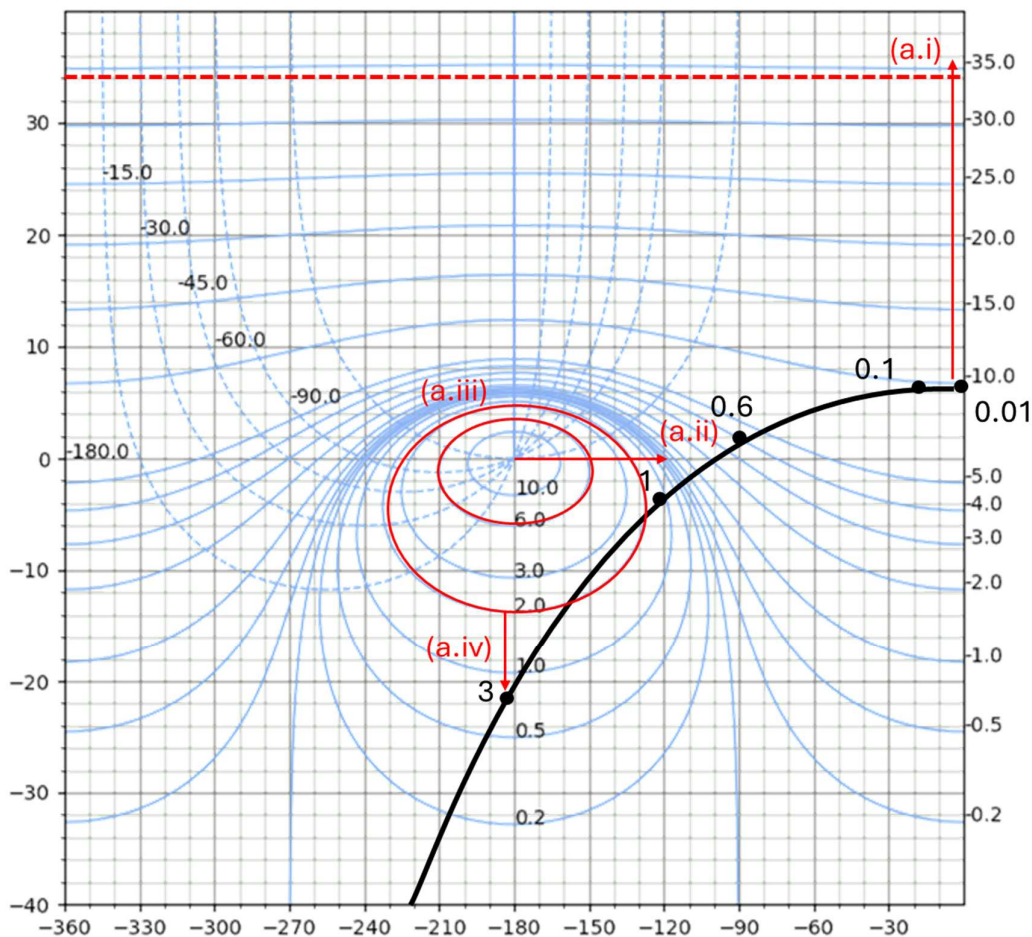
- i. Tracking of constant setpoints with less than 2% accuracy.
 - ii. Overshoot of less than 10%.
 - iii. Sensitivity of less than 6 dB at all frequencies.
 - iv. Sensitivity of less than 2 dB at 3 rad/s.
-
- a) Sketch the frequency response of the system on the inverse Nichols chart provided and indicate boundaries for each of these specifications. [5 marks]
 - b) For each specification, determine the range of gains for which it will be satisfied. You may state your answers in decibels. [4 marks]
 - c) What is the best possible accuracy with respect to position setpoints without violating specifications ii-iii? [2 marks]
 - d) Is the accuracy specification achievable for this system even if the other specifications are ignored? If not, give the best achievable accuracy. [3 marks]





Solution:

- a) To get the frequency on the inverse Nichols chart, I plotted the frequencies at the boundaries of the decades, some intermediate ones that looked easy to plot (on gridlines) and the frequency of interest at 3 rad/s.
 - i. From the FV theorem applied to the error TF, less than 2% error for positions requires $\frac{1}{1+KG(0)} \leq 0.02 \therefore KG(0) \geq 49$ i.e. 33.8 dB (≈ 34)
 - ii. From the percentage overshoot formula, less than 10% means a damping ratio of greater than 0.6. At the desired accuracy (i.e. a closed-loop gain of 0.98), that would equate to a phase margin of $0.6 \left(\frac{100}{0.98} \right) = 61^\circ$
 - iii. The global sensitivity specification amounts to keeping the whole response out of the 6dB S circle.
 - iv. The marked frequency of 3 rad/s must be kept out of the 2 dB S circle.



- b) For this question, you need to calculate how much the gain can increase by before the specification is either violated or achieved.
 - i. The difference between the current DC gain and the required gain is $K \geq 34 - 6 = 28$ dB.
 - ii. From the Bode plot, the point where the response has a phase of 60 degrees is around 1 rad/s, where the magnitude is about -4 dB. So, the gain can increase by 4 dB before this is violated, so $K \leq 4$ dB.

- iii. The response is around 8 dB below the -6 circle, so $K \leq 8$ dB.
 - iv. The magnitude at 3 rad/s is about -22 dB, putting it about 8 dB below the 2 dB S circle. So, $K \leq 8$ dB.
- c) The smallest increase in gain that will cause any of specifications ii-iii to be violated is 4 dB, so we are looking at the accuracy for this increase. The gain of the system is currently 6 dB, so this will raise it to 10 dB, or 3.1. This gives closed-loop error of $\frac{1}{1+3.1} = 24\%$ i.e. 76% accuracy.
- d) The gain margin of the system is only around 20 dB, so increasing the gain by 28 dB will make the closed loop unstable, therefore, the accuracy specification cannot be achieved. If we consider the accuracy at the maximum gain of 20 dB, the new DC gain of the system will be 26 dB, or 20. This gives a closed-loop error of $\frac{1}{1+20} = 0.05$ i.e. 95% accuracy.

Marking notes: tutors, please take into account that the students are working from hand-drawn plots so please allow uncertainty of ± 2 dB when evaluating whether an answer is correct.